BUSINESS ANALYSIS

Functional Specification Document

Functional Requirements Specification

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Functional Specification Document

Laundry Service Shop App — “WashLoop”

Field	Detail
Document Title	Functional Requirements Specification — Laundry Service Shop App
Document ID	BS-001
Version	1.0 (Draft for review)
Date	15 June
Author(s)	Senior Business Analyst (Functional Spec Writer Agent)
Sponsor	Laundry Service Shop Owner / Operations Manager
Related Area	Customer Operations, Logistics (Pickup), Payments, Mobile & Web
Document Status	IN PROGRESS — awaiting sign-off
Platform	Native iOS, Native Android, Lightweight Web Admin Portal
Budget / Timeline	$15,000 – $18,000 · 12 weeks

Document Approvers & Reviewers

Name	Role	Approver / Reviewer	Approval / Review Date	Version
Shop Owner	Business Sponsor	Approver	TBD	1.0
Operations Manager	Operational Lead	Approver	TBD	1.0
Lead Developer	Technical Lead	Reviewer	TBD	1.0
QA Lead	Head of UAT	Approver	TBD	1.0

Reference Documents

- Interview Q&A record (Product Overview, KPIs, Scope, NFRs, Roles, Integrations, Constraints, Dependencies & Risks)
- Requirements Summary (summarizer_agent output)
- Existing POS system API documentation (to be validated — see Assumptions)

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1. Introduction

1.1 Overview

WashLoop is a customer-facing and operations-facing application for a residential laundry service shop. It digitises the shop’s end-to-end service flow: a customer schedules a pickup, a driver collects the laundry, shop staff process the items (wash & fold, dry cleaning, or ironing), the customer tracks the order in real time, and the customer pays for the service through Stripe, Apple Pay or Google Pay. SMS and email notifications keep the customer informed at every stage.

The solution consists of three deliverables:

1. **Customer mobile app** — native iOS and native Android.
2. **Operations workflow** — driver and sorting/front-desk staff update order status from the app.

3. **Web admin portal** — a lightweight browser-based console for order management, driver assignment, pricing/inventory sync with the existing POS, and operational reporting.

The shop currently handles approximately **57 orders per day (~400 orders per week)** for roughly **200 active customers per week**, with an average order value of **$30**, generating approximately **$1,710 gross revenue per day / $51,000 per month**. Peak order windows are **11:00–14:00** and **16:00–18:00**, reaching **10–12 orders per hour**.

1.2 Objectives

#	Objective	Measure of Success
O1	Replace manual/phone-based order intake with self-service scheduling	≥ 70% of orders booked through the app by end of beta
O2	Provide transparent, real-time order status to customers	Status updated within 2 seconds of the staff action; customer-visible
O3	Reduce pickup no-shows and scheduling errors	On-time pickup rate ≥ 90% (target — to be ratified by owner)
O4	Increase repeat usage	Repeat-order rate ≥ 40% of active customers per month (target — to be ratified)
O5	Digitise payment collection	≥ 95% payment success rate; PCI-DSS compliant
O6	Give management operational visibility	Daily order, revenue and driver-utilisation dashboards available

1.3 Business Benefits

- Fewer manual touchpoints and phone calls per order.
- Consistent pricing and inventory sourced from the existing POS system.
- Higher customer trust through status transparency and notification trails.
- Capacity to absorb peak-window demand without front-desk bottlenecks.
- Data foundation for future loyalty, delivery and routing features (v2 candidates).

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2. Scope

2.1 In Scope (v1 / MVP)

Area	Capability
Customer app	Account registration/login, pickup scheduling (same-day and next-day), service-type selection (wash & fold, dry clean, ironing), order creation, order status tracking, in-app payment, notification settings
Payments	Stripe integration, Apple Pay and Google Pay via Stripe; order-level refunds initiated by admin
Notifications	SMS (Twilio) and email notifications for order confirmation, pickup confirmation, status change and completion

Area	Capability
Operations	Driver pickup assignment, pickup-completed marking, staff status updates through a defined status lifecycle
Web admin portal	Order queue management, driver assignment, pricing and inventory sync from POS, basic operational and revenue reporting
Integrations	Stripe, Twilio, existing POS API (pricing and inventory), Apple/Google payment sheets
Platforms	iOS app, Android app, responsive web portal (Chrome/Edge/Safari)
Data & security	Customer PII storage, PCI-DSS compliant payment token handling, role-based access control

2.2 Out of Scope (v1)

Note: The items below were raised during the requirements interview but **not explicitly confirmed** by the business owner. They are recorded here as **assumed out of scope for v1** and must be ratified before development starts. See Section 7.

Item	Status in v1	Rationale
Home return / delivery of cleaned laundry	Assumed OUT (customer collection only)	Not confirmed by owner
Loyalty / referral / points programme	OUT	Deferred to v2
Insurance, damage or lost-item claims workflow	OUT	Exception handling not defined
Driver GPS routing / live vehicle tracking	OUT	Deferred to v2
Self-service drop-off at shop	OUT	Not part of confirmed flow
Multi-location / franchise support	OUT	Single shop assumed
Advanced analytics & BI	OUT	Basic reporting only
Offline-first driver mode with sync	Assumed OUT (limited resilience required)	Not confirmed
Subscriptions / recurring weekly plans	OUT	Deferred

2.3 Systems & Platforms

System	Type	Notes
iOS app	Native (iOS 15+)	App Store distribution
Android app	Native (Android 10+)	Play Store distribution
Web admin portal	Responsive web app	Lightweight; admin + front-desk use
Backend API	Cloud-hosted REST API	≥ 15 req/s peak capacity
Stripe	External payment gateway	Cards, Apple Pay, Google Pay
Twilio	External SMS/OTP gateway	Notification dispatch
Existing POS	External system	Pricing & inventory sync

3. Functional Requirements

Priority scheme: MoSCoW — **Must** (v1 mandatory), **Should** (v1, deferrable within a sprint), **Could** (nice-to-have), **Won't** (explicitly deferred to v2 or beyond). **Source column** identifies the stakeholder who raised the requirement.

3.1 Ordering & Pickup Scheduling

ID	Requirement	Priority	Source
FR-001	The system shall allow a registered customer to create an order requesting a laundry pickup.	Must	Owner
FR-002	The system shall offer both **same-day** and **next-day** pickup options at order creation.	Must	Owner
FR-003	The system shall let the customer select one or more service types: Wash & Fold, Dry Cleaning, Ironing, or a mixed load.	Must	Owner
FR-004	The system shall capture the pickup address, contact phone number, preferred pickup window, and estimated load (bags / items).	Must	Owner
FR-005	The system shall validate the pickup address against the defined service area before confirming the order.	Must	Ops Manager
FR-006	The system shall reject or re-schedule a pickup request when the selected window	Should	Ops Manager

ID	Requirement	Priority	Source
	exceeds available driver capacity.		
FR-007	The system shall suggest the next available pickup window when the requested window is unavailable.	Should	Ops Manager

ID	Requirement	Priority	Source
FR-008	The system shall allow a customer to cancel a scheduled pickup **before collection**, and shall record the cancellation reason.	Must	Owner
FR-009	The system shall allow a customer to reschedule a pickup window subject to capacity.	Should	Ops Manager
FR-010	The system shall record an audit trail (user, timestamp, change) for every order lifecycle change.	Must	Ops Manager

3.2 Driver & Collection Workflow

ID	Requirement	Priority	Source
FR-011	The admin portal shall display a pickup queue of all scheduled collection stops for a given day.	Must	Ops Manager
FR-012	The admin portal shall allow a driver to be assigned to each pickup stop.	Must	Ops Manager
FR-013	The driver app shall show only the stops assigned to that driver for their shift.	Must	Driver lead
FR-014	The driver shall be able to confirm a collection with a single action, moving the order status to **Collected**.	Must	Driver lead
FR-015	The driver shall be able to record the number of bags/items collected and any observed condition notes.	Should	Ops Manager
FR-016	The driver shall be able to log a **Missed Pickup** exception with a reason code (customer unavailable, access issue, route delay).	Must	Ops Manager
FR-017	The system shall automatically notify the customer when a pickup is missed and offer a reschedule.	Must	Owner

ID	Requirement	Priority	Source
FR-018	The driver app shall retain locally captured pickup actions when connectivity is lost and synchronise them on reconnection. *(Assumed requirement — see Assumption A6)*	Could	Ops Manager

3.3 Shop Processing & Order Status Lifecycle

ID	Requirement	Priority	Source
FR-019	The system shall support the order status lifecycle: **Created → Scheduled → Pickup Assigned → Collected → Received → Sorting → In Processing → Ready → Completed / Cancelled**.	Must	Ops Manager
FR-020	Shop staff shall be able to mark an order as **Received** at the shop and link it to the customer.	Must	Front desk
FR-021	Sorting staff shall be able to categorise the load by service type (Wash & Fold / Dry Cleaning / Ironing) and record quantity.	Must	Shop staff
FR-022	Shop staff shall be able to update processing progress (washing, drying, ironing, finished).	Must	Shop staff
FR-023	The system shall flag orders as **Delayed** when they exceed the expected turnaround window.	Should	Ops Manager
FR-024	The system shall allow staff to log a **damaged item** or **missing item** exception against an order, with a description.	Must	Front desk

ID	Requirement	Priority	Source
FR-025	The system shall support a front-desk lookup of an order by order number, customer name, phone number, or bag tag.	Must	Front desk
FR-026	Front-desk staff shall be able to verify, release and mark an order **Completed** when the customer collects it.	Must	Front desk
FR-027	The customer shall be able to view the current order status, service type, and processing stage in real time from the app.	Must	Owner

3.4 Payments

ID	Requirement	Priority	Source
FR-028	The system shall accept payment	Must	Owner

ID	Requirement	Priority	Source
	through **Stripe** (card), **Apple Pay**, and **Google Pay**.		
FR-029	The system shall calculate the order total from service types and pricing synchronised from the POS API, including applicable taxes and service fees.	Must	Owner
FR-030	The system shall handle payment either at scheduling or before collection release, and shall clearly display the outstanding balance.	Must	Front desk
FR-031	The payment flow shall complete in **≤ 2 seconds** under normal load.	Must	Owner
FR-032	All card data shall be handled via Stripe payment tokens; the application shall **not** store raw card numbers.	Must	Technical lead
FR-033	The system shall support a partial or full refund initiated by an authorised admin or front-desk user. *(Workflow detail unconfirmed — see Assumption A3)*	Should	Owner

ID	Requirement	Priority	Source
FR-034	The system shall record payment status (Paid, Pending, Failed, Refunded) on every order and expose it to the admin portal.	Must	Ops Manager
FR-035	The system shall retry or fail gracefully a payment attempt without creating a duplicate transaction.	Must	Technical lead
FR-036	The system shall issue a payment receipt to the customer by email and/or SMS link.	Should	Owner

3.5 Notifications

ID	Requirement	Priority	Source
FR-037	The system shall send **SMS** notifications through **Twilio** and **email** notifications for key order events: confirmation, pickup confirmation, status change, ready-for-collection, and completion.	Must	Owner
FR-038	Notifications shall be dispatched within **5 seconds** of the triggering status change.	Must	Owner
FR-039	The notification service shall sustain approximately **1 message/second** in normal operation and **5 messages/second** at peak.	Must	Technical lead
FR-040	Customers shall be able to choose	Must	Owner

ID	Requirement	Priority	Source
	their preferred notification channel (SMS, email, or both) and opt out of marketing messages.		
FR-041	The system shall log notification delivery status (sent / failed) and retry failed SMS delivery once.	Should	Ops Manager
FR-042	The system shall send an in-app push notification alongside SMS for app-installed customers.	Could	Owner

3.6 Web Admin Portal & Reporting

ID	Requirement	Priority	Source
FR-043	The admin portal shall present a filtered, sortable order queue with status, date, service type, and payment status.	Must	Owner
FR-044	The admin portal shall allow manual order creation and edit by front-desk staff.	Must	Front desk
FR-045	The admin portal shall support driver assignment and reassignment.	Must	Ops Manager
FR-046	The admin portal shall synchronise **prices and inventory/consumables** with the existing POS API at least daily, with on-demand refresh.	Must	Owner
FR-047	The admin portal shall display basic operational reports: daily orders, revenue, service-type mix, on-time pickup rate, and cancellations.	Must	Owner
FR-048	The admin portal shall expose KPI dashboards for the metrics defined in Section 5.3.	Should	Owner
FR-049	The admin portal shall be usable on desktop and tablet browsers (Chrome, Edge, Safari).	Must	Technical lead

ID	Requirement	Priority	Source
FR-050	The admin portal shall enforce role-based access (Admin, Ops Manager, Front Desk) and log privileged actions.	Must	Technical lead

3.7 Accounts, Access & Integrations

ID	Requirement	Priority	Source
FR-051	The system shall allow customer registration via email or mobile number with OTP verification.	Must	Technical lead
FR-052	The system shall allow customers to save address and payment-method details for faster repeat ordering.	Should	Owner
FR-053	The system shall provide role-specific interfaces: customer app, driver view, staff view, admin portal.	Must	Ops Manager
FR-054	The system shall integrate with the **POS API** for pricing and inventory data. *(Readiness unconfirmed — see Risk R1)*	Must	Owner
FR-055	The system shall integrate with Stripe, Twilio, Apple Pay and Google Pay using documented APIs and stored credentials in a secure secrets store.	Must	Technical lead
FR-056	The system shall provide an exportable order history (CSV) for the reporting period requested by the owner.	Could	Owner
FR-057	The system shall retain order and payment records for a minimum of 12 months for audit purposes.	Should	Owner

4. Non-Functional Requirements

4.1 Performance & Capacity

ID	Requirement	Target
NFR-001	API throughput during peak windows (11:00–14:00 and 16:00–18:00)	**≥ 15 requests/second**
NFR-002	Customer checkout / order-placement response time	**≤ 2 seconds**
NFR-003	Admin portal screen load (order-status board, queues)	**≤ 1 second**
NFR-004	Order status change reflected to the customer	**≤ 5 seconds**
NFR-005	SMS / email notification dispatch after a status change	**≤ 5 seconds**
NFR-006	Concurrent users supported during peak	**6–8 concurrent users** (customers + 3–4 front-desk staff)
NFR-007	Spike headroom (local event, promotion burst)	**10–15 simultaneous sessions**, a 20% safety margin above normal peak
NFR-008	Payment gateway throughput without latency degradation	**≥ 30 transactions/hour** (≈ 1–2 calls/second)
NFR-009	Notification dispatch rate	1 msg/s normal; **5 msg/s** peak
NFR-010	Throughput baseline	≥ 1 order processed every 30 seconds during peak

4.2 Availability & Reliability

ID	Requirement	Target
NFR-011	System availability (customer app, API, admin portal)	**99.9% uptime** (≈ 8 hours downtime per year maximum)
NFR-012	Graceful degradation	Order intake and status viewing must continue (read-only) if Stripe or Twilio is temporarily unavailable
NFR-013	Failed notification handling	Automatic single retry; failures logged and surfaced in admin portal
NFR-014	Web tier sizing	Minimum **2 CPU cores / 4 GB RAM**, with auto-scaling triggered when concurrency exceeds 12 sessions
NFR-015	Backup & recovery	Daily automated database backup; recovery point objective ≤ 24 hours; recovery time objective ≤ 4 hours

4.3 Security & Compliance

ID	Requirement	Detail
NFR-016	**PCI-DSS** compliance for all payment handling; card data never stored on shop infrastructure	Stripe tokenisation required
NFR-017	Customer PII (name, address, phone, payment tokens) encrypted at rest and in transit	TLS 1.2+; AES-256 at rest
NFR-018	Role-based access control with least-privilege permissions across customer, driver, staff, manager and admin roles	Enforced server-side
NFR-019	Session management with token expiry and secure login	MFA recommended for admin accounts
NFR-020	Full audit logging of order status changes, refunds and price overrides	Retained ≥ 12 months
NFR-021	Compliance with applicable privacy legislation (GDPR/ CCPR-equivalent) for consent, data access and deletion requests	To be confirmed by legal review

4.4 Usability & Accessibility

ID	Requirement	Detail
NFR-022	New customers should be able to place their first order in **≤ 3 steps** and in under 2 minutes	Design target
NFR-023	Interface usable in low-light shop environments with large touch targets for staff terminals	Design target
NFR-024	Support accessibility standards (WCAG 2.1 AA) on the web portal and platform-level accessibility on mobile	Design target
NFR-025	Staff require **≤ 1 hour** of training to perform order entry and status updates	Ops Manager
NFR-026	Clear status labels and colour-coded progress indicators for non-technical customers	Design target

4.5 Maintainability, Compatibility & Environment

ID	Requirement	Detail
NFR-027	iOS app compatible with iOS 15+; Android app compatible with Android 10+	Compatibility
NFR-028	Web portal responsive on desktop (≥ 1280px) and tablet (≥ 768px)	Compatibility
NFR-029	Configuration of service types, prices, and pickup windows changeable without code release	Maintainability
NFR-030	Environment separation: Development, Staging (beta rollout), Production	Maintainability

ID	Requirement	Detail
NFR-031	Monitoring & alerting on uptime, API latency and notification queue depth	Maintainability

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5. User Roles & Personas

5.1 Roles and Permissions Matrix

Role	Channel	Can Create Order	Can Assign Driver	Can Update Status	Can Take Payment	Can Issue Refund	Can View Reports	Can Manage Pricing/POS Sync
Customer	iOS / Android app	Yes	No	No	Yes	No	No (own orders only)	No
Driver	Driver view (mobile)	No	No	Pickup only (Collected / Missed)	No	No	No (own stops only)	No
Sorting / Shop Staff	Staff view	No	No	Processing stages (Received, Sorting, In Processing, Ready)	No	No	No	No

Role	Channel	Can Create Order	Can Assign Driver	Can Update Status	Can Take Payment	Can Issue Refund	Can View Reports	Can Manage Pricing/POS Sync
Front-Desk Staff	Web portal / staff view	Yes (walk-in / phone orders)	No	Received, Completed, exception logging	Yes	Yes (with approval *(open item)*	Limited (daily ops)	No
Shop Operations Manager	Web portal	Yes	Yes	All stages	Yes	Yes	Yes (ops KPIs)	No
Admin / Owner	Web portal	Yes	Yes	All stages	Yes	Yes	Yes (full)	Yes

Note on permissions: The interview did **not** definitively confirm whether drivers or sorting staff update status, who issues refunds, or which users can access the web portal. The matrix above

is the **recommended v1 model** and must be ratified by the owner (see Assumptions A3–A5 and Open Items O2–O3).

5.2 Personas

P1 — Residential Customer ("Maya")

- 32, professional, uses laundry pickup 2× per week (one wash & fold, occasional dry clean).
- Goals: save time, know exactly where her laundry is, pay without a queue.
- Frustrations: unclear turnaround times, no visibility of what has been collected.
- Technical comfort: high — comfortable with app-based services and digital payments.

P2 — Driver ("Ravi")

- Works 3–4 drivers across shifts, 15–20 stops per shift.
- Goals: clear list of stops, one-tap confirmation, minimal phone calls to the shop.
- Frustrations: wrong addresses, customers not home, paper-based status updates.
- Technical comfort: moderate — needs a simple, high-contrast mobile interface.

P3 — Sorting / Shop Staff ("Aisha")

- Processes ~57 orders/day, 2–3 staff during peak.
- Goals: unambiguous service-type labels, simple status updates, flag damaged/missing items.
- Frustrations: mismatch between what the customer requested and what arrives.
- Technical comfort: low-to-moderate — requires short training and simple screens.

P4 — Front-Desk Staff ("Tom")

- 3–4 employees handling walk-ins, payments, collections and queries.
- Goals: fast order lookup (<1 second), clear payment status, easy refund/credit handling.
- Frustrations: duplicate orders, unclear outstanding balances.
- Technical comfort: moderate.

P5 — Shop Operations Manager ("Dana")

- Manages driver allocation, peak-window capacity, exceptions and daily reporting.
- Goals: see bottlenecks early, measure on-time pickup and driver utilisation.
- Technical comfort: high.

P6 — Owner / Administrator ("Sam")

- Sets pricing, approves refunds, reviews revenue and KPIs, owns the POS relationship.
- Goals: one dashboard for orders, revenue and service mix; trust in payment compliance.
- Technical comfort: moderate.

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6. Assumptions & Constraints

6.1 Assumptions (documented, unconfirmed — require ratification)

ID	Assumption	Impact if wrong
A1	v1 ends at **customer collection at the shop**; home return delivery is **out of scope**.	Major scope change; route management and delivery staffing needed
A2	Loyalty, insurance/ damage claims, driver GPS routing, self drop-off and subscriptions are **v2+**.	Scope creep against $15–18k budget
A3	Refunds are possible and are initiated by admin or approved front-desk staff; a partial-refund workflow is required.	Payment and reconciliation rework
A4	Drivers update only pickup-related status; sorting staff update processing status; front desk completes orders.	Role/permission redesign
A5	Admin and Operations Manager have web portal access; front-desk staff have	Access-control and licensing cost

ID	Assumption	Impact if wrong
	a limited portal view.	
A6	Drivers operate with **occasional** connectivity loss; a lightweight local-cache/ sync model is sufficient (full offline mode not required).	Additional development effort

ID	Assumption	Impact if wrong
A7	The existing **POS API is documented and accessible** with adequate rate limits.	Integration delay; possible replacement of pricing source
A8	Stripe and Twilio accounts are (or will be) provisioned with production credentials before development week 3.	Payment/notification milestones slip
A9	Single shop location, single currency, single tax regime.	Multi-site/tax complexity
A10	KPI target values (on-time pickup %, status accuracy %, repeat-order rate) are the recommended defaults in Section 5.3 and will be ratified by the owner.	Reporting definitions change
A11	~200 active customers per week and ~400 orders per week are the planning volumes for capacity sizing.	Infrastructure under/over-provisioning

6.2 Constraints

ID	Constraint	Detail
C1	**Budget**	Total design, development and testing must remain within **$15,000 – $18,000**
C2	**Timeline**	12 weeks total: 2 weeks design → 6 weeks development → 2 weeks QA → 2 weeks beta rollout
C3	**Platforms**	Native iOS and native Android apps plus a lightweight web portal — no single web-only app
C4	**Existing systems**	Pricing and inventory must come from the **existing POS**; a new pricing engine is not in budget
C5	**Third-party services**	Stripe and Twilio are mandated; alternative providers are not in scope

ID	Constraint	Detail
C6	**Staffing**	3–4 drivers and 2–3 processing staff at peak; 3–4 front-desk users on the system
C7	**App store governance**	iOS/Android store approval timelines are outside the team's control
C8	**Compliance**	PCI-DSS is mandatory; raw card data cannot be stored
C9	**Small team**	No dedicated DevOps or security team is assumed; cloud-managed services preferred

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7. Open Items Requiring Business Owner Input

The requirements interview covered all eight subject areas, but the following seven items were **not answered explicitly**. They are captured as assumptions above and must be closed before development begins.

#	Open Item	Current assumption	Needed by
O1	Home return delivery in v1?	No — customer collection only	Week 2 (design sign-off)
O2	Refund workflow (who authorises, partial vs full, time limit)	Admin approves; full/partial supported	Week 3
O3	Who updates order status: driver vs sorting staff	Driver = pickup; staff = processing	Week 2
O4	POS API documented/accessible today?	Yes	Week 1 (technical spike)
O5	KPI target values (on-time pickup %, status accuracy %, repeat rate)	Recommended defaults in §5.3	Week 4
O6	Driver offline behaviour required?	Light local caching with sync	Week 3
O7	Stripe / Twilio / app-store accounts provisioned?	To be provisioned by week 3	Week 3
O8	Uptime SLA confirmation	99.9% (provided by owner)	Confirmed

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8. Use Cases & User Stories

8.1 Assumptions Governing the Use Cases

- **Collection at the shop is assumed for v1.** Home return delivery is an open item (O1) and is not modelled as confirmed v1 functionality.
- Customers can schedule **same-day or next-day pickup**.
- Pickup is confirmed by an assigned driver in the app.
- Shop staff receive, sort, process and release orders for customer collection at the shop.

- Customers are notified by SMS (Twilio) and email; payments run through Stripe / Apple Pay / Google Pay.
- Non-functional targets applied throughout: 99.9% uptime, checkout ≤ 2 s, admin screens ≤ 1 s, notifications ≤ 5 s, API ≥ 15 req/s, 6–8 concurrent users with a 10-session safety margin, PCI-DSS compliant.

8.2 UC-001 — Customer Schedules a Pickup

Actors: Customer, Front-desk staff (fallback), System, Operations Manager, Driver **Goal:** Create a scheduled pickup order for the next available same-day or next-day window.

Preconditions: Customer account exists or is identified by phone; pickup address is within the service area; operating hours and driver capacity are known.

Main flow: 1. Customer opens the app (or contacts the front desk). 2. Customer selects **same-day** or **next-day** pickup. 3. Customer enters address, preferred time window and estimated load. 4. Customer selects service mix (wash & fold, dry cleaning, ironing, or mixed). 5. System validates identity, address, window and capacity. 6. System creates the order with status **Scheduled**. 7. System places the stop in the pickup queue for driver assignment. 8. Operations Manager (or automated rule) assigns a driver. 9. System sends SMS/email confirmation via Twilio. 10. Customer receives a tracking reference/link.

Alternative / exception flows: - **Cancellation before pickup** → status becomes **Cancelled**; if payment was taken, the refund/credit workflow is triggered; customer is notified. - **Invalid address or window** → validation error displayed; customer or front desk corrects and re-submits. - **No capacity** → next available window suggested; customer accepts or contacts the shop. - **Capacity exceeded mid-shift** → Operations Manager reassigns or reschedules and the customer is notified.

Postconditions: Order exists in the admin pickup queue; stop visible to the assigned driver; customer can track status; confirmation notification delivered.

Acceptance criteria: Same-day/next-day selection available; address and window validated before confirmation; order appears in the admin queue within 1 second; confirmation SMS/email sent within 5 seconds; peak load of 10–12 orders/hour handled with API ≥ 15 req/s.

8.3 UC-002 — Driver Confirms Collection

Actors: Driver, Customer, System, Operations Manager, Front desk **Goal:** Confirm physical collection of laundry and move the order to **Collected**.

Preconditions: Pickup scheduled and assigned; driver has the route list; customer has laundry ready.

Main flow: 1. Driver logs into the driver view. 2. Driver reviews assigned stops for the shift. 3. Driver arrives at the location and opens the stop. 4. Driver taps **Collected**. 5. Driver optionally records number of bags/items, service notes and any condition observations. 6. System updates status from *Scheduled* to *Collected*. 7. System notifies the customer and the front desk. 8. Order enters the shop receiving/sorting queue.

Alternative / exception flows: - **Missed pickup** → driver selects a reason code (customer unavailable, access issue, route delay); status becomes **Missed Pickup**; Operations Manager reviews; customer notified and offered a reschedule. - **Connectivity loss** → driver action is cached locally

and synchronised on reconnect; sync delay beyond threshold raises an exception flag *(open item O6)*. - **Uncertain customer/stop** → driver does not confirm; exception logged; front desk contacted. - **Damaged or missing items noted at pickup** → condition logged; exception created; front desk contacts customer before processing.

Postconditions: Status **Collected**; exception logged if any; route status updated; customer notified; order in shop receiving queue.

Acceptance criteria: One-tap confirmation; status visible to all within 5 seconds; missed-pickup exception visible in the admin queue; offline actions synchronise with an audit trail; driver cannot confirm a stop whose address fails validation.

8.4 UC-003 — Shop Receives, Sorts and Categorises Laundry

Actors: Front-desk staff, Sorting staff, System, Operations Manager **Goal:** Identify the received load, assign service types, and record exceptions.

Preconditions: Order status is **Collected**; laundry physically at the shop; order identifiable by number, customer, phone or bag tag.

Main flow: 1. Staff receive the laundry and identify the order. 2. Staff mark the order **Received**. 3. Sorting staff separate items and assign service categories. 4. Staff record quantity/load estimate. 5. Staff flag missing, damaged or unusual items. 6. System moves status to **Sorting** / *Ready for Processing*. 7. Order enters the processing queue.

Alternative / exception flows: - **Damaged item found** → condition recorded; exception created; front desk seeks customer confirmation; processing may be held. - **Missing item reported** → expected vs received compared; exception raised; customer notified; refund/credit may be triggered. - **Unidentifiable load** → orphan item recorded; lookup attempted; escalation if unresolved. - **Service mix mismatch** → actual service type recorded with reason; price adjustment applied from POS pricing.

Postconditions: Status **Received/Sorting**; service types recorded; exceptions logged; audit trail written.

Acceptance criteria: Staff can mark received and sort in-system; mixed service types supported; exceptions visible to front desk and Operations Manager; admin queue reflects status within 1 second.

8.5 UC-004 — Process Order and Update Progress Status

Actors: Sorting/shop staff, System, Customer, Front desk **Goal:** Track processing stages and give the customer real-time visibility.

Preconditions: Order received and sorted; service type assigned.

Main flow: 1. Staff begin processing; status becomes **In Processing**. 2. Wash & fold: wash → dry → fold stages recorded. 3. Dry cleaning: cleaning stage recorded. 4. Ironing: pressing stage recorded. 5. Quality check performed. 6. Staff mark order **Ready for Collection**. 7. System notifies the customer. 8. Front desk prepares the order for handover.

Alternative / exception flows: - **Processing delay** → status **Delayed**; customer notified; Operations Manager reviews queue impact. - **Rework required** → status remains In Processing; customer

contacted if delay is significant. - **Missing item discovered** → exception logged; order held; refund/credit workflow triggered. - **Special handling requested** → note recorded and flagged for staff.

Postconditions: Status reflects current stage; ready orders visible for collection; delays and exceptions surfaced; customer notified.

Acceptance criteria: Full status progression supported; notifications within 5 seconds of each change; ~57 orders/day and 400 orders/week sustained; peak windows of 10–12 orders/hour handled.

8.6 UC-005 — Customer Tracks Order Status

Actors: Customer, Front desk, System **Goal:** Provide a live view of pickup, processing, payment and collection readiness.

Preconditions: Order exists; customer has an app account or tracking link.

Main flow: 1. Customer opens the app or the tracking link from the SMS/email. 2. System retrieves the current status. 3. Customer sees pickup schedule, collection, sorting, processing stage, payment status and readiness. 4. Customer contacts the front desk if the status is unclear. 5. When the order is **Ready**, the customer visits the shop to collect.

Alternative / exception flows: - **Tracking link unavailable** → front desk looks up the order by name, phone, order number or bag tag. - **Status lag** → last known status displayed with a “last updated” timestamp. - **Payment outstanding** → tracking page shows balance and a checkout link.

Postconditions: Customer and front desk see the same status; payment state visible before collection.

Acceptance criteria: Status accurate within 5 seconds of change; front-desk lookup returns in ≤ 1 second; supports 6–8 concurrent users plus spike headroom.

8.7 UC-006 — Notify Customers on Status Change (SMS/Email)

Actors: System, Twilio, Customer, Front desk **Goal:** Communicate every material order event promptly and reliably.

Preconditions: Order status change occurred; valid customer contact details; Twilio credentials configured.

Main flow: 1. System detects a status change. 2. Notification template selected (confirmation, pickup, processing, ready, completion, exception). 3. System composes the message with order reference. 4. Message dispatched via Twilio SMS and/or email within 5 seconds. 5. Delivery outcome recorded (sent/failed). 6. Failed SMS retried once; persistent failure logged to the admin portal.

Alternative / exception flows: - **Twilio outage** → message queued; email fallback sent; queue depth alerted. - **Invalid/unreachable number** → delivery marked failed; front desk prompted to confirm contact details. - **Opt-out** → only transactional (non-marketing) messages sent.

Postconditions: Customer informed; delivery log written; queue drains within 5 s/msg capacity.

Acceptance criteria: ≤ 5 s dispatch; 1 msg/s sustained, 5 msg/s peak; retries logged; opt-out respected.

8.8 UC-007 — Customer Pays for the Order

Actors: Customer, System, Stripe, Apple Pay / Google Pay, Front desk **Goal:** Collect payment securely and promptly.

Preconditions: Order total calculated from POS-synced pricing; payment method available.

Main flow: 1. Customer reviews the order total (service fees, taxes). 2. Customer selects Stripe card, Apple Pay or Google Pay. 3. System creates a Stripe payment token. 4. Payment authorised and recorded; status becomes **Paid**. 5. Receipt sent by email/SMS link. 6. Order flagged as releasable for collection.

Alternative / exception flows: - **Payment declined** → clear error shown; alternate method offered; order saved as *Pending*; no duplicate transaction created. - **Gateway timeout** → system retries once with idempotency key; on failure the customer is informed and the order held. - **Front-desk cash/ POS payment** → staff record payment status manually, reconciled against POS.

Postconditions: Payment recorded; order unlocked for collection; receipt issued.

Acceptance criteria: Checkout ≤ 2 seconds; PCI-DSS compliant with no raw card data stored; ≥ 30 transactions/hour without latency; payment success rate ≥ 95%.

8.9 UC-008 — Refund or Credit Handling

Actors: Admin/Owner, Front-desk staff, Operations Manager, System, Customer **Goal:** Resolve cancellations, missing/damaged items and over-charges financially. *(Workflow detail is an open item — O2.)*

Preconditions: Order exists with a recorded payment; a qualifying exception has been logged.

Main flow: 1. Exception recorded (cancelled, damaged, missing item). 2. Authorised user initiates refund or credit against the Stripe transaction. 3. Refund amount (full or partial) approved by Admin/ Owner. 4. System applies the refund and updates payment status to **Refunded**. 5. Customer notified by SMS/email. 6. Refund logged in the audit trail and reporting.

Alternative / exception flows: - **Partial refund** → credit issued against future orders instead of cash refund. - **Refund denied** → reason recorded; customer notified. - **Refund exceeds original charge** → system blocks the action and escalates to Owner.

Postconditions: Financial record accurate; audit trail complete; customer informed.

Acceptance criteria: Only authorised roles can refund; refunds reflect the original amount; audit trail captures approver and timestamp; cancellation/refund resolution target ≤ 24 hours.

8.10 UC-009 — Admin Manages the Order Queue and Assigns Drivers

Actors: Operations Manager, Admin/Owner, System **Goal:** Keep pickups balanced across drivers and shift capacity.

Preconditions: Scheduled orders exist; driver roster available for the shift.

Main flow: 1. Admin opens the order queue in the web portal. 2. Admin filters by date, window, status and service type. 3. Admin assigns or reassigns drivers to stops. 4. Drivers see the updated stops in their app. 5. Admin monitors on-time performance and delays.

Alternative / exception flows: - **Capacity exceeded** → admin reschedules stops or extends windows; affected customers notified. - **Driver absent** → stops reassigned; notification sent to affected customers.

Postconditions: Balanced driver workload; queue status current; assignment logged.

Acceptance criteria: Queue renders in ≤ 1 second; assignment propagates to driver view within 5 seconds; role permissions enforced.

8.11 UC-010 — POS Price and Inventory Synchronisation

Actors: Admin/Owner, System, POS API **Goal:** Ensure pricing and consumable stock in the app match the shop's POS. *(POS readiness is an open item — O4.)*

Preconditions: POS API credentials available; price catalogue defined.

Main flow: 1. System calls the POS API for service prices and stock levels. 2. Data mapped to app service types (wash & fold, dry clean, ironing). 3. Sync executed daily and on demand. 4. Admin reviews sync status and any mapping errors. 5. Updated prices applied to new orders.

Alternative / exception flows: - **POS API unavailable** → last known prices used; warning displayed to admin; orders still created. - **Price mismatch detected** → flagged for manual confirmation before publication.

Postconditions: Pricing consistent; stock levels current; sync audit logged.

Acceptance criteria: Daily sync succeeds; manual refresh available; graceful fallback to cached prices; no order blocked by POS outage.

8.12 UC-011 — Operational Reporting and KPI Review

Actors: Owner, Operations Manager, System **Goal:** Measure service performance and business outcomes.

Preconditions: Orders processed; KPI definitions agreed *(target values are open item O5)*.

Main flow: 1. System aggregates order, payment, status and notification data. 2. Dashboard displays daily orders, revenue, service mix, on-time pickup %, cancellations, refunds and notification success. 3. Owner/manager reviews trends and takes action. 4. Report exported as CSV on request.

Acceptance criteria: Reports accurate against transactional data; refreshed at least daily; KPI definitions documented.

8.13 UC-012 — Customer Completes Collection at the Shop

Actors: Customer, Front-desk staff, System **Goal:** Hand over the finished laundry and close the order.

Preconditions: Order status **Ready**; payment settled or balance agreed.

Main flow: 1. Customer arrives and presents the order reference. 2. Front desk verifies the order and payment status. 3. Staff release the laundry and mark the order **Completed**. 4. System sends completion notification. 5. Optional satisfaction prompt sent.

Alternative flows: Customer requests later collection → order remains Ready with a note; item shortage at handover → exception and refund workflow.

Acceptance criteria: Order closed with timestamp; notification sent within 5 seconds; order removed from active queue.

8.14 Use Case → User Story Mapping

Use Case	User Story	Priority	Related FR
UC-001	As a **customer**, I want to schedule a same-day or next-day pickup, so that I do not have to visit the shop.	Must	FR-001–FR-007
UC-001	As a **customer**, I want to choose my service type, so that I get exactly the service I need.	Must	FR-003
UC-001	As a **customer**, I want to cancel a pickup before collection, so that I am not charged for a service I did not use.	Must	FR-008
UC-002	As a **driver**, I want to confirm a pickup with one tap, so that I spend no time on manual reporting.	Must	FR-014, FR-015
UC-002	As a **driver**, I want to record a missed pickup with a reason, so that the shop can reschedule quickly.	Must	FR-016, FR-017
UC-002	As a **driver**, I want my pickup actions to work when the network is down, so	Could	FR-018

Use Case	User Story	Priority	Related FR
	that I can still record collections.		

Use Case	User Story	Priority	Related FR
UC-003	As a **sorting staff member**, I want to record what arrived and its service type, so that processing is accurate.	Must	FR-020, FR-021
UC-003	As a **front-desk staff member**, I want to flag damaged or missing items, so that the customer can be contacted before work starts.	Must	FR-024
UC-004	As a **shop staff member**, I want to update processing progress, so that the customer always knows where the order is.	Must	FR-022, FR-023
UC-005	As a **customer**, I want real-time order status, so that I know when to collect my laundry.	Must	FR-027, FR-025
UC-006	As a **customer**, I want SMS and email updates, so that I do not have to check the app.	Must	FR-037–FR-041
UC-006	As a **customer**, I want to choose my notification channel, so that I only receive messages I want.	Must	FR-040

Use Case	User Story	Priority	Related FR
UC-007	As a **customer**, I want to pay securely with Apple Pay, Google Pay or a card, so that payment is fast and safe.	Must	FR-028–FR-032
UC-007	As a **front-desk staff member**, I want to see whether an order is paid,	Must	FR-034

Use Case	User Story	Priority	Related FR
	so that I can release it confidently.		
UC-008	As an **owner**, I want to issue refunds for cancelled or problematic orders, so that customers are treated fairly.	Should	FR-033
UC-009	As an **operations manager**, I want to assign drivers to stops, so that pickups happen on time.	Must	FR-012, FR-045
UC-010	As an **owner**, I want prices and stock to sync with my POS, so that pricing is always correct.	Must	FR-046, FR-054
UC-011	As an **owner**, I want daily reporting on orders, revenue and on-time pickups, so that I can make decisions.	Must	FR-047, FR-048

Use Case	User Story	Priority	Related FR
UC-012	As a **customer**, I want a smooth collection experience when I pick up my laundry, so that the service feels complete.	Must	FR-026
UC-001	As a **customer**, I want saved addresses and payment details, so that repeat orders take less than 2 minutes.	Should	FR-052

8.15 Definition of Done (applies to every backlog item)

An item is complete only when it is: 1. Implemented and integrated with the required external system (Stripe, Twilio or POS). 2. Tested against the stated acceptance criteria, including peak-load behaviour. 3. Reviewed for security (RBAC, PCI, PII handling). 4. Documented in the specification and release notes. 5. Verified by the nominated stakeholder (source of the requirement).

8.16 Key Performance Indicators & Success Metrics

Target values marked *proposed* were not confirmed by the business owner during the interview (open item **O5**) and must be ratified before the QA phase.

KPI	Definition	Proposed launch target	Owner
Active customers	Unique customers placing ≥ 1 order in a week	200 / week	Owner
Order volume	Orders created per day / week	57 / day · 400 / week	Ops Manager
Service mix	Wash & Fold / Dry Clean / Ironing split	60% / 30% / 10%	Owner
Average order value	Gross revenue ÷ orders	≈ $30	Owner
Gross revenue	Daily / weekly / monthly	$1,710 / $12,000 / $51,000	Owner
On-time pickup rate	Pickups completed within the promised window ÷ scheduled pickups	**≥ 90%** *(proposed)*	Ops Manager
Missed-pickup rate	Missed pickups ÷ scheduled pickups	≤ 5% *(proposed)*	Ops Manager
Order-status accuracy	Statuses matching physical reality at spot checks	≥ 98% *(proposed)*	Ops Manager

KPI	Definition	Proposed launch target	Owner
Notification delivery rate	Delivered ÷ attempted notifications	≥ 97% *(proposed)*	Technical lead
Payment success rate	Successful ÷ attempted transactions	≥ 95% *(proposed)*	Owner
Repeat-order rate	Customers placing ≥ 2 orders in a month	≥ 40% *(proposed)*	Owner
Driver utilisation	Stops completed ÷ stops assigned	≥ 85% *(proposed)*	Ops Manager
Cancellation/refund resolution time	Exception logged → resolution	≤ 24 hours *(proposed)*	Front desk
System uptime	Measured availability	99.9% (confirmed)	Technical lead
Peak checkout latency	Order placement response time	≤ 2 s (confirmed)	Technical lead
App adoption	Orders placed via app ÷ total orders	≥ 70% by end of beta *(proposed)*	Owner

KPI reporting ownership: the Shop Operations Manager produces the weekly KPI pack; the Owner/Admin ratifies thresholds and reviews monthly trends. Reporting is generated from the admin portal dashboards (FR-048).

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9. Glossary

9.1 Service & Operational Terms

Term	Definition
Wash & Fold	Laundry service in which items are washed, dried and folded for customer collection.
Dry Cleaning	Professional chemical/specialty cleaning for garments that cannot be standard-washed.
Ironing	Finishing service in which washed garments are pressed to remove creases.
Service Type	Classification of the requested service: Wash & Fold, Dry Cleaning, Ironing, or a mixed load.
Order	A customer-initiated request for laundry service, including service type, pickup details, timing, payment status and collection information.
Pickup Window	The time range in which a driver is scheduled to collect laundry from a customer's location.
Same-Day Pickup	Collection on the same calendar day the order is created.
Next-Day Pickup	Collection on the next business day after the order is created.
Home Return Delivery	Delivery of finished laundry back to the customer's address — **open item; assumed out of scope for v1**.
Self Drop-Off	Customer delivers laundry directly to the shop — out of scope for v1.

Term	Definition
Order Status Lifecycle	The ordered sequence of statuses an order moves through: Created → Scheduled → Pickup Assigned → Collected → Received → Sorting → In Processing → Ready → Completed / Cancelled.
Created	Order submitted and recorded, not yet scheduled.
Scheduled	Order assigned a pickup schedule; collection not yet performed.
Pickup Assigned	A driver has been allocated to the collection stop.
Collected	Laundry physically collected by the driver.
Received	Laundry physically received and logged at the shop.
In Processing	Order actively undergoing washing, dry cleaning, ironing or sorting.
Ready	Processing complete; order prepared for customer collection.
Completed	Order collected by the customer, closing the service cycle.
Cancelled	Order terminated before completion, by customer, staff or system exception.
Exception	Any deviation from the normal flow (missed pickup, damaged item, missing item, delay).

Term	Definition
Turnaround Time	Elapsed time from collection to the order being Ready.

9.2 Roles

Term	Definition
Customer	End user who requests laundry services, schedules pickup, pays, and collects completed laundry.
Driver	Shop staff member responsible for collecting laundry from customer locations and confirming collection in the system.
Sorting / Shop Staff	Staff who receive, sort, categorise and process laundry items and update processing status.
Front-Desk Staff	Staff handling customer interaction, order entry, lookup, payment collection, refunds/credits and order handover.
Shop Operations Manager	Operational lead responsible for scheduling, driver assignment, capacity management, exceptions and daily reporting.
Admin / Owner	Business owner with full system access: pricing, POS integration, permissions, refund approvals and reporting.

9.3 Systems, Integrations & Technical Terms

Term	Definition
POS API	Application programming interface connecting WashLoop to the shop's existing point-of-sale system for pricing, inventory and transaction data.
Stripe	Payment processing platform used to accept card, Apple Pay and Google Pay transactions.
Apple Pay	Digital wallet payment method for Apple devices, integrated via Stripe.
Google Pay	Digital wallet payment method for Android/Google Pay-enabled devices, integrated via Stripe.
Twilio	Communication service used to send SMS notifications, confirmations and alerts.
SMS Notification	Text message sent to a customer or staff member about an order event.
Push Notification	In-app alert delivered through the mobile application.
Payment Token	Secure identifier representing a customer's payment method, used instead of raw card data.
PCI-DSS	Payment Card Industry Data Security Standard governing the handling and protection of card data.
Web Admin Portal	Browser-based console for order management, driver assignment, POS sync and reporting.

Term	Definition
Real-Time Status Tracking	Live visibility of order progress, pickup state, processing stage and collection readiness.

Term	Definition
RBAC	Role-Based Access Control — permissions granted according to a user's role.
Audit Trail	Chronological record of who changed what, and when, within the system.
Idempotency Key	Token ensuring a retried payment does not create a duplicate transaction.
Auto-Scaling	Automatic provisioning of additional application instances when concurrency exceeds a defined threshold.

9.4 Metrics & Non-Functional Terms

Term	Definition
Uptime SLA	Agreed minimum system availability over a period, expressed as a percentage (here 99.9%).
Peak Concurrency	Maximum number of simultaneous users, sessions or API requests the system must support without degradation.
Response Time	Elapsed time from a request being received to it being processed.
Latency	Delay introduced by network transmission, integration calls or system processing.
Throughput	Number of requests or transactions the system can process per unit of time.
On-Time Pickup Rate	Percentage of scheduled pickups completed within the agreed window.
Order-Status Accuracy	Degree to which displayed statuses match the actual physical state of an order.
Repeat-Order Rate	Percentage of customers placing further orders after a completed service (retention measure).
Driver Utilisation	Effectiveness of driver deployment relative to assigned stops and available capacity.
Notification Delivery Rate	Percentage of sent notifications successfully delivered to the recipient.

Term	Definition
Payment Success Rate	Percentage of attempted transactions completing without error or timeout.
Graceful Degradation	Maintaining core functionality (read-only status, queued notifications) when a third-party service is unavailable.

9.5 Project & Documentation Terms

Term	Definition
MVP / v1	First release containing only the core capabilities needed to launch and validate demand.
MoSCoW	Prioritisation method: **M**ust, **S**hould, **C**ould, **W**on't.
Acceptance Criteria	Conditions a feature must satisfy to be accepted as complete.
Definition of Done	Agreed criteria confirming a deliverable is implemented, integrated, tested and documented.
Assumption	A premise accepted as true for planning purposes although not yet formally confirmed.
Constraint	A limitation restricting the solution (budget, time, technology, legal, operational).
Dependency	A condition where one component or process relies on another to proceed.
Risk	A potential event that could negatively affect delivery, quality, security or operations.
Open Item	An unresolved question or unconfirmed decision requiring stakeholder input.
Beta Rollout	Controlled release to a limited user group before full production launch.

9.6 Acronyms

Acronym	Meaning
MVP	Minimum Viable Product
v1 / v2	Version 1 / Version 2
SLA	Service Level Agreement
NFR	Non-Functional Requirement
FR	Functional Requirement
UC	Use Case

Acronym	Meaning
POS	Point of Sale
API	Application Programming Interface
PCI-DSS	Payment Card Industry Data Security Standard
SMS	Short Message Service
OTP	One-Time Password
RBAC	Role-Based Access Control
KPI	Key Performance Indicator
PII	Personally Identifiable Information
CSV	Comma-Separated Values
WCAG	Web Content Accessibility Guidelines
TLS	Transport Layer Security

Acronym	Meaning
CPU	Central Processing Unit
RAM	Random Access Memory
TPS	Transactions Per Second
RPO / RTO	Recovery Point Objective / Recovery Time Objective
MoSCoW	Must, Should, Could, Won't

Appendix A — Traceability Summary

Requirement area	Confirmed by interview?	Notes
Product Overview & core flow	Yes	Pickup → process → track → notify → pay
Service types	Yes	Wash & Fold 60%, Dry Clean 30%, Ironing 10%
Volumes & revenue	Yes	57 orders/day, ~400/week, AOV ~$30, ~$51k/month
Peak windows & concurrency	Yes	11:00–14:00 & 16:00–18:00; 10–12 orders/hr; 6–8 users, 10-session headroom
Performance targets	Yes	≥15 req/s, ≤2 s checkout, ≤1 s admin, ≤5 s notifications
Uptime SLA	Yes	99.9%
Compliance	Yes	PCI-DSS required
Integrations	Yes	Stripe, Apple/Google Pay, Twilio, POS API
Budget & timeline	Yes	$15–18k; 12 weeks
Home return delivery	**No**	Assumed out of scope (A1)

Requirement area	Confirmed by interview?	Notes
Refund workflow	**No**	Assumed admin-approved (A3)
Driver vs staff status permissions	**No**	Recommended model in §5.1 (A4)
POS API readiness	**No**	Assumed available (A7)
KPI target values	**No**	Proposed targets in §8.16 (A10)
Driver offline behaviour	**No**	Light caching assumed (A6)
Stripe/Twilio/app-store provisioning	**No**	Must be provisioned by week 3 (A8)

Appendix B — Version History

Version	Date	Author	Change
0.1	Interview close	BA	Initial draft from interview Q&A
1.0	This document	BA	Full functional specification: scope, FRs, NFRs, roles, assumptions, use cases, glossary
1.1	TBD	BA	Update after owner ratifies open items O1–O7

10. Dependencies & Risks

10.1 External Dependencies

#	Dependency	Description	Status	Mitigation / Action
D1	**POS API**	Pricing and inventory/ consumables sync (FR-046, FR-054)	Unconfirmed — documentation and access not verified	Run a 2-day technical spike in Week 1; if unavailable, fall back to a manually maintained price catalogue cached in the app
D2	**Stripe account & API keys**	Card, Apple Pay and Google Pay processing	Not confirmed as provisioned	Provision sandbox + production keys by Week 3; PCI scope limited to tokenisation
D3	**Twilio account**	SMS notification dispatch	Not confirmed as provisioned	Provision and verify sending limits before Week 4; add email fallback

#	Dependency	Description	Status	Mitigation / Action
D4	**Apple App Store / Google Play approval**	Distribution of native apps	Timeline outside team control	Submit beta builds early (Week 9) and build review buffers into the 12-week plan
D5	**Driver onboarding & devices**	3–4 drivers with compatible Android/ iOS devices and training	Operational readiness assumed	Onboard and train drivers during beta; provide printable route sheets as fallback
D6	**Staff training**	3–4 front-desk and 2–3 processing staff need role training	Assumed ≤ 1 hour training	Produce quick-start guides; involve staff in UAT
D7	**Cloud hosting / auto-scaling**	Web tier sized at 2 CPU / 4 GB with scaling at 12 sessions	To be selected	Choose managed cloud (auto-scale) to fit within budget and C9 constraint
D8	**Customer contact data quality**	Valid mobile numbers and emails for notifications	Assumed collected at registration	Validate at registration; flag undeliverable contacts in the admin portal

10.2 Risk Register

#	Risk	Likelihood	Impact	Mitigation
R1	POS API undocumented or rate-limited, delaying pricing sync	Medium	High	Early technical spike; cached-price fallback; manual price override in admin portal
R2	Payment integration complexity (Stripe + Apple/ Google Pay) slips the schedule	Medium	High	Implement Stripe first (cards), add digital wallets in the second development sprint
R3	App store approval extends beyond the 12-week timeline	Medium	Medium	Submit for review in Week 9; keep a web/PWA fallback for early beta customers
R4	Scope creep (delivery, loyalty, GPS routing, claims) exceeds $15–18k budget	High	High	Freeze v1 scope at sign-off; log deferred items as v2 backlog; change-control board for any additions
R5	Undefined exception handling (missed pickup,	High	Medium	Ratify open items O1–O3 before Week 3; include exception flows in UAT scripts

#	Risk	Likelihood	Impact	Mitigation
	damage, refunds) creates operational gaps			
R6	SMS delivery rate limits or Twilio outage reduce notification reliability	Low–Medium	Medium	Queue and retry messages; email fallback; monitor queue depth and alert on backlog

#	Risk	Likelihood	Impact	Mitigation
R7	Peak-window concurrency exceeds sizing (15–20 orders/hr spikes)	Medium	Medium	Auto-scaling above 12 sessions; 20% capacity headroom; load testing in QA at ≥ 15 req/s
R8	Driver connectivity gaps cause status out-of-sync	Medium	Medium	Local caching with sync and audit trail; define sync-failure exception handling
R9	Staff adoption resistance to new status-update workflow	Medium	Medium	Short training, simple UI, phased rollout, and champion users per shift
R10	Compliance exposure if card data is mishandled	Low	High	Stripe tokenisation only; no raw card storage; security review before production
R11	Ambiguous role permissions lead to incorrect status updates or unauthorised refunds	Medium	Medium	Enforce RBAC server-side; approval workflow for refunds; audit logging
R12	KPI thresholds not agreed, making “success” unmeasurable	Medium	Medium	Ratify proposed targets (§8.16) with the owner before QA

10.3 Risk-Adjusted Timeline Note

The 12-week plan (2 design → 6 development → 2 QA → 2 beta) is achievable **only if** the POS API spike (D1) and payment gateway provisioning (D2) complete inside the first 3 weeks. If either slips, the recommended trade-off is to reduce scope rather than extend the budget — specifically by deferring FR-042 (push notifications), FR-056 (CSV export) and FR-048 (KPI dashboards), all of which are *Should/Could* priority.

Document Approval

Role	Name	Approver / Reviewer	Date	Version Approved
Head of UAT	QA Lead	Approver	TBD	1.0
Head of Business Change	Operations Manager	Approver	TBD	1.0
Business Sponsor	Shop Owner	Approver	TBD	1.0

Role	Name	Approver / Reviewer	Date	Version Approved
Technical Lead	Lead Developer	Reviewer	TBD	1.0

This specification is ready for review. Closure of the eight open items in Section 8 (O1–O8) is a prerequisite for design sign-off and development start.

11. Index of Sections (Reader Navigation)

#	Section	Contents
1	Introduction	Overview, objectives, business benefits
2	Scope	In scope (v1), out of scope (v1), systems & platforms
3	Functional Requirements	FR-001 → FR-057 (7 requirement groups, MoSCoW priority)
4	Non-Functional Requirements	NFR-001 → NFR-031 (performance, availability, security, usability, maintainability)
5	User Roles & Personas	Permissions matrix + 6 personas
6	Assumptions & Constraints	A1–A11 assumptions; C1–C9 constraints
7	Open Items	O1–O8 requiring owner ratification
8	Use Cases & User Stories	UC-001 → UC-012, story map, Definition of Done, KPIs
9	Glossary	Service terms, roles, systems, metrics, project terms, acronyms
10	Dependencies & Risks	D1–D8 dependencies; R1–R12 risk register
11	Appendices	Traceability summary, version history, approvals

Cross-Reference Corrections

- **KPI / success metrics** are defined in **Section 8.16**. Where FR-048 and Assumption A10 refer to "Section 5.3", read it as **Section 8.16**.
- **Open items** are listed in **Section 7** (O1–O8). The Document Approval note referencing "Section 8" should read **Section 7**.

Next Steps for the Business Owner

1. Ratify the eight open items (O1–O8) — in particular **home return delivery (O1)**, **refund workflow (O2)** and **role permissions (O3)**, which materially affect scope and cost.
2. Confirm the **POS API technical spike** (D1/R1) in Week 1 — this is the single largest schedule risk to the 12-week plan.
3. Approve the **proposed KPI thresholds** in §8.16 so that QA can be written against measurable targets.
4. Confirm **Stripe, Twilio and app-store account provisioning** (A8/D2/D3/D4) before development week 3.

Once these are closed, the specification moves to version 1.1 and the design phase (weeks 1–2) can begin within the **$15,000–$18,000** budget envelope.

Process Diagram

End-to-End Order Lifecycle Process Flow (UC-001 to UC-012)

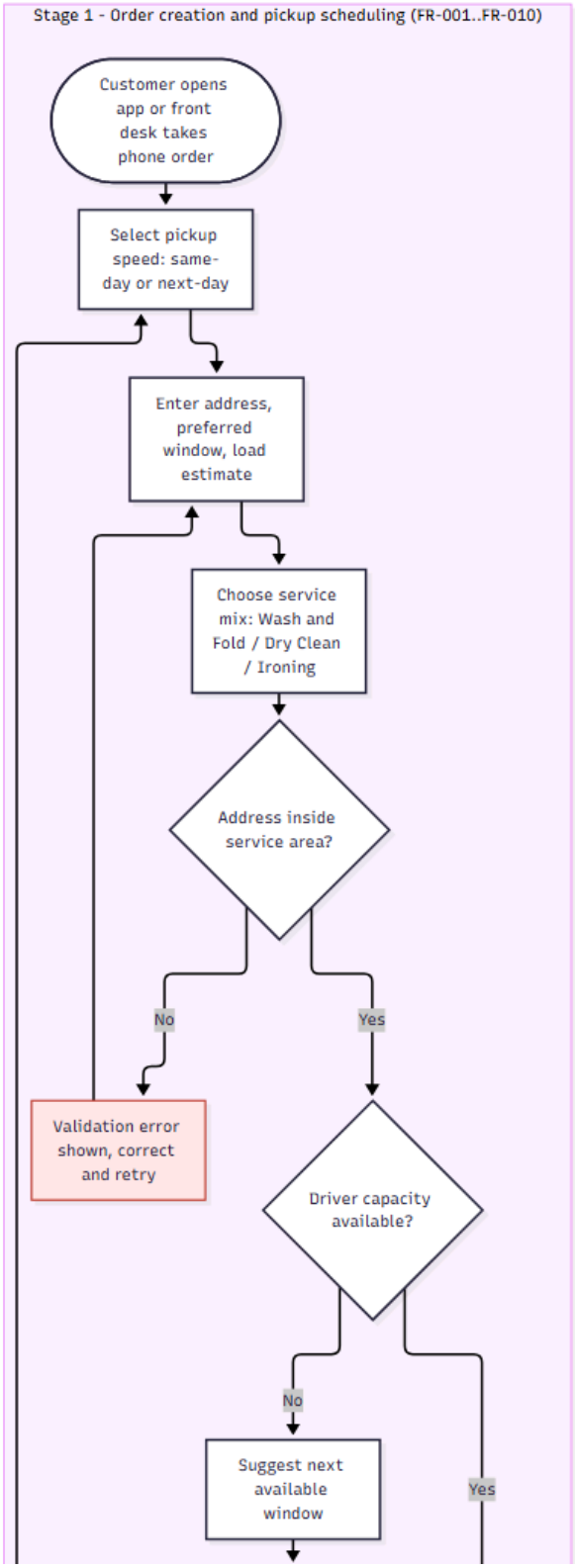


Figure 1: Process Diagram (part 1 of 8)

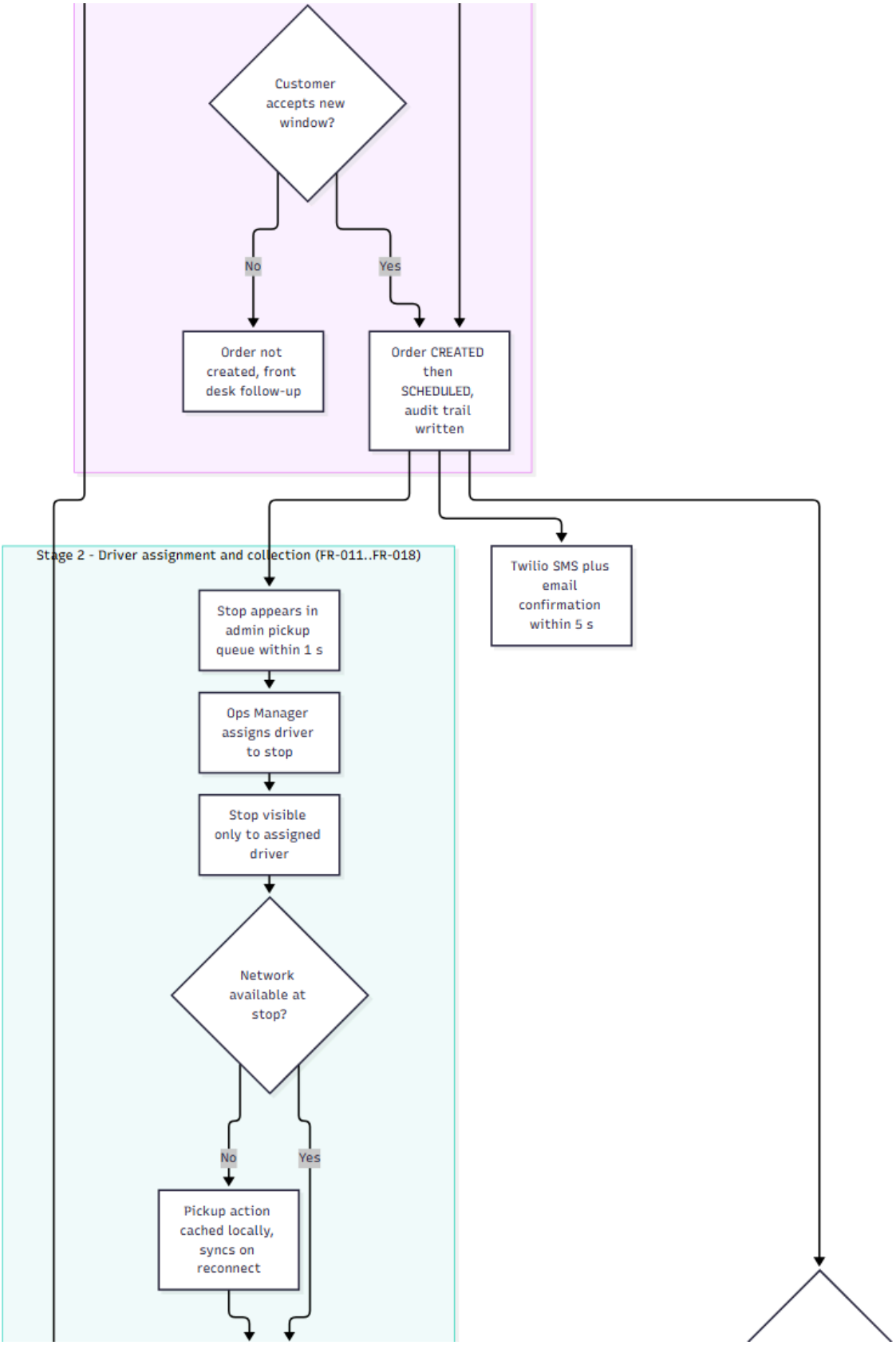


Figure 2: Process Diagram (part 2 of 8)

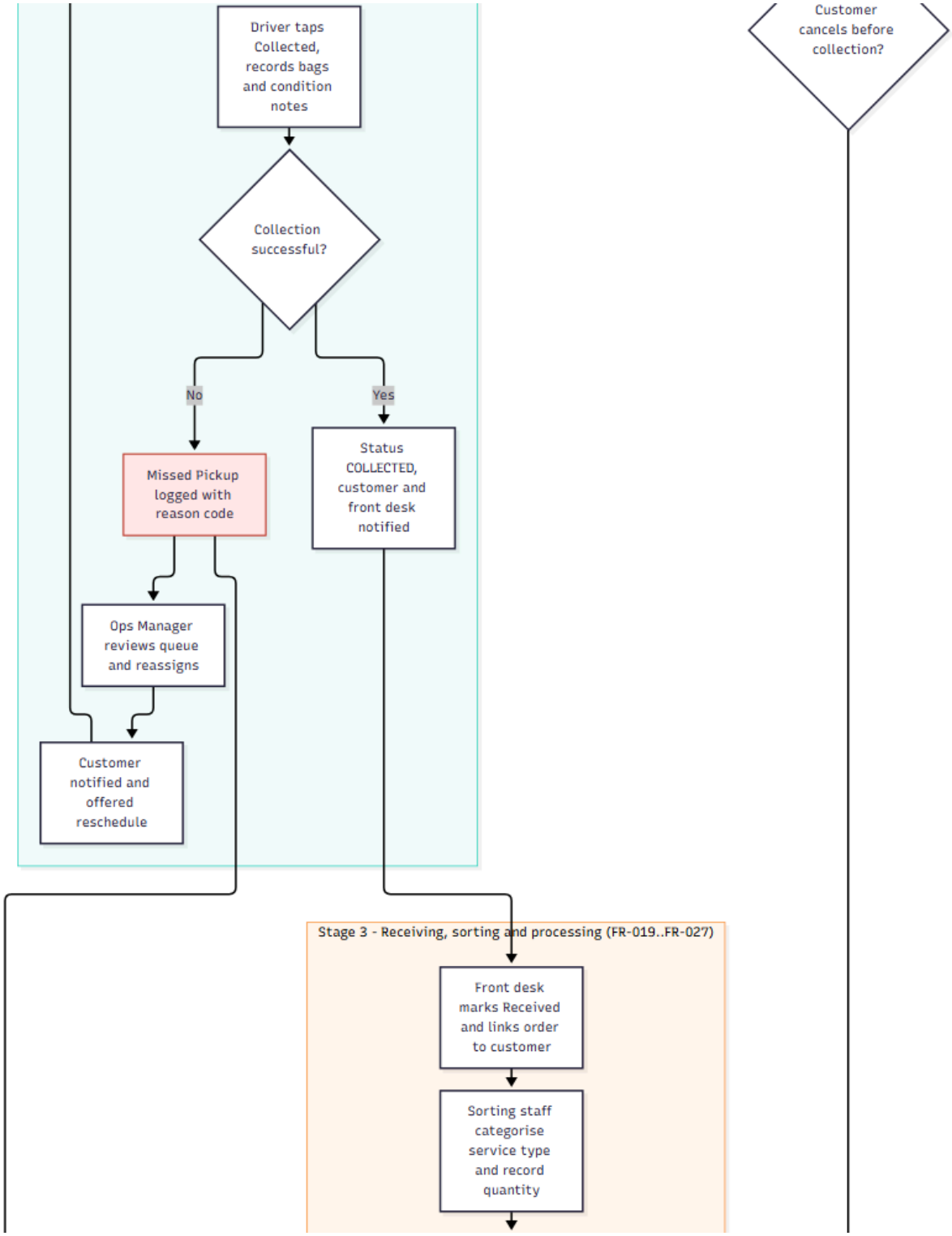


Figure 3: Process Diagram (part 3 of 8)

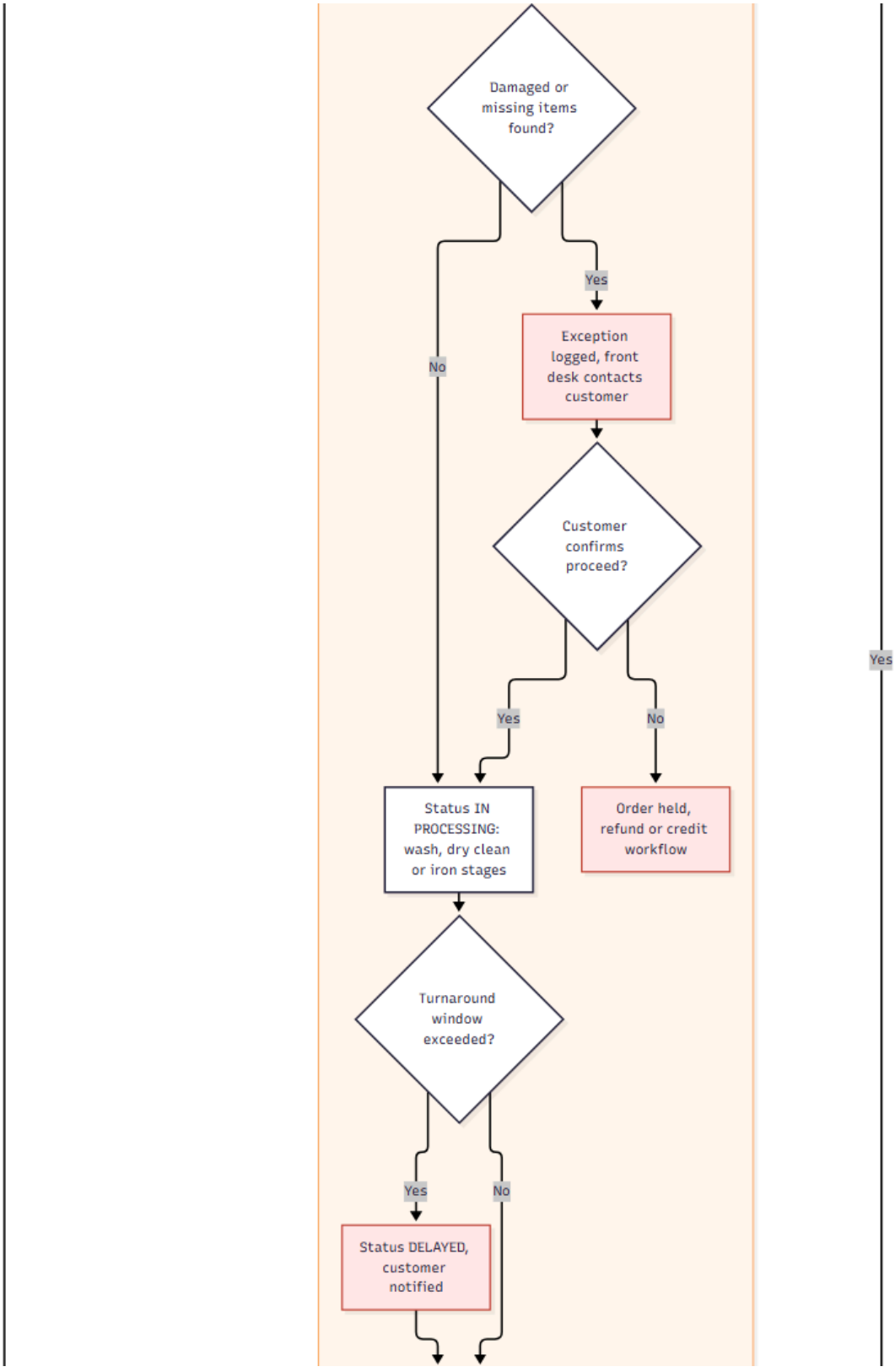


Figure 4: Process Diagram (part 4 of 8)

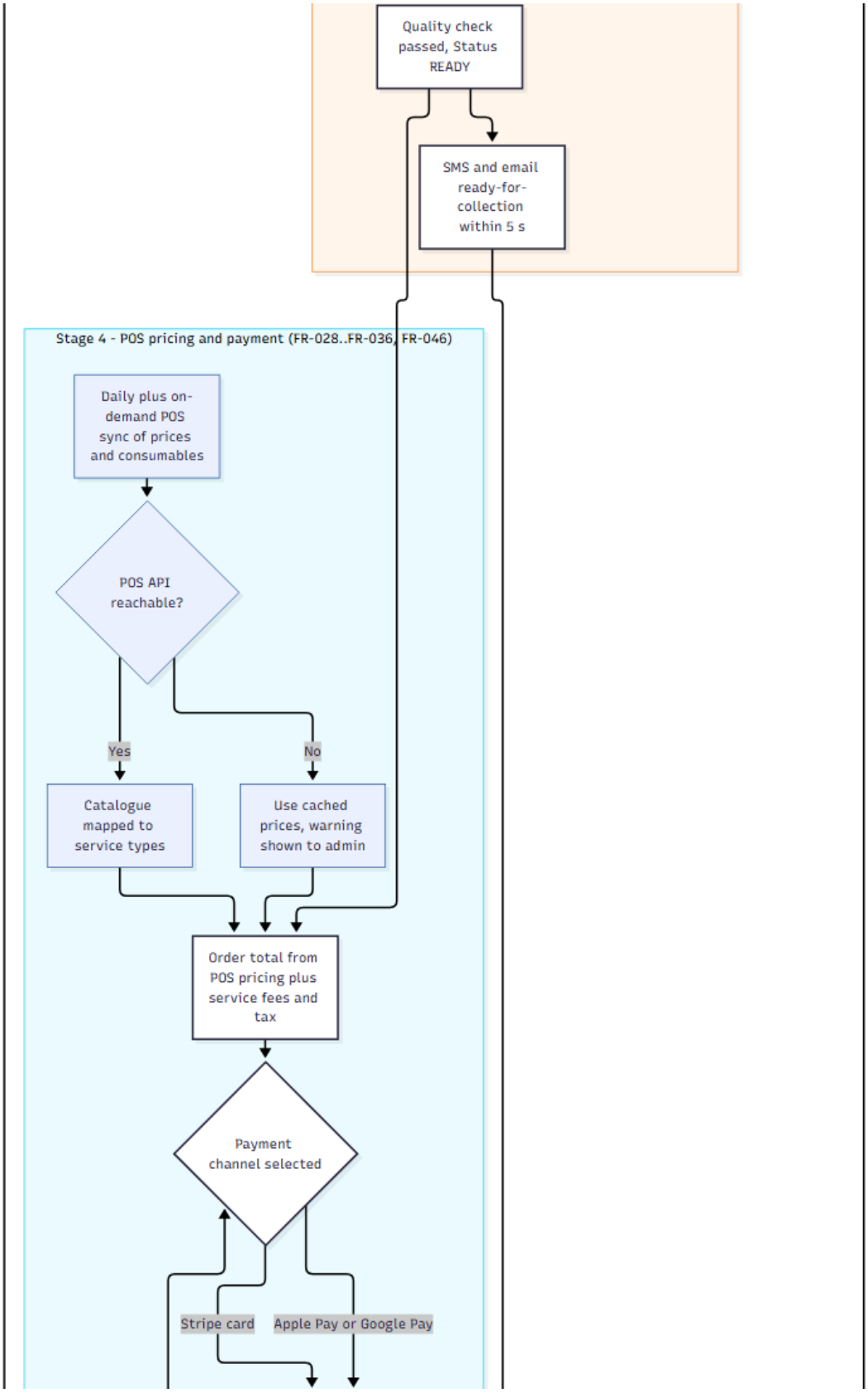


Figure 5: Process Diagram (part 5 of 8)

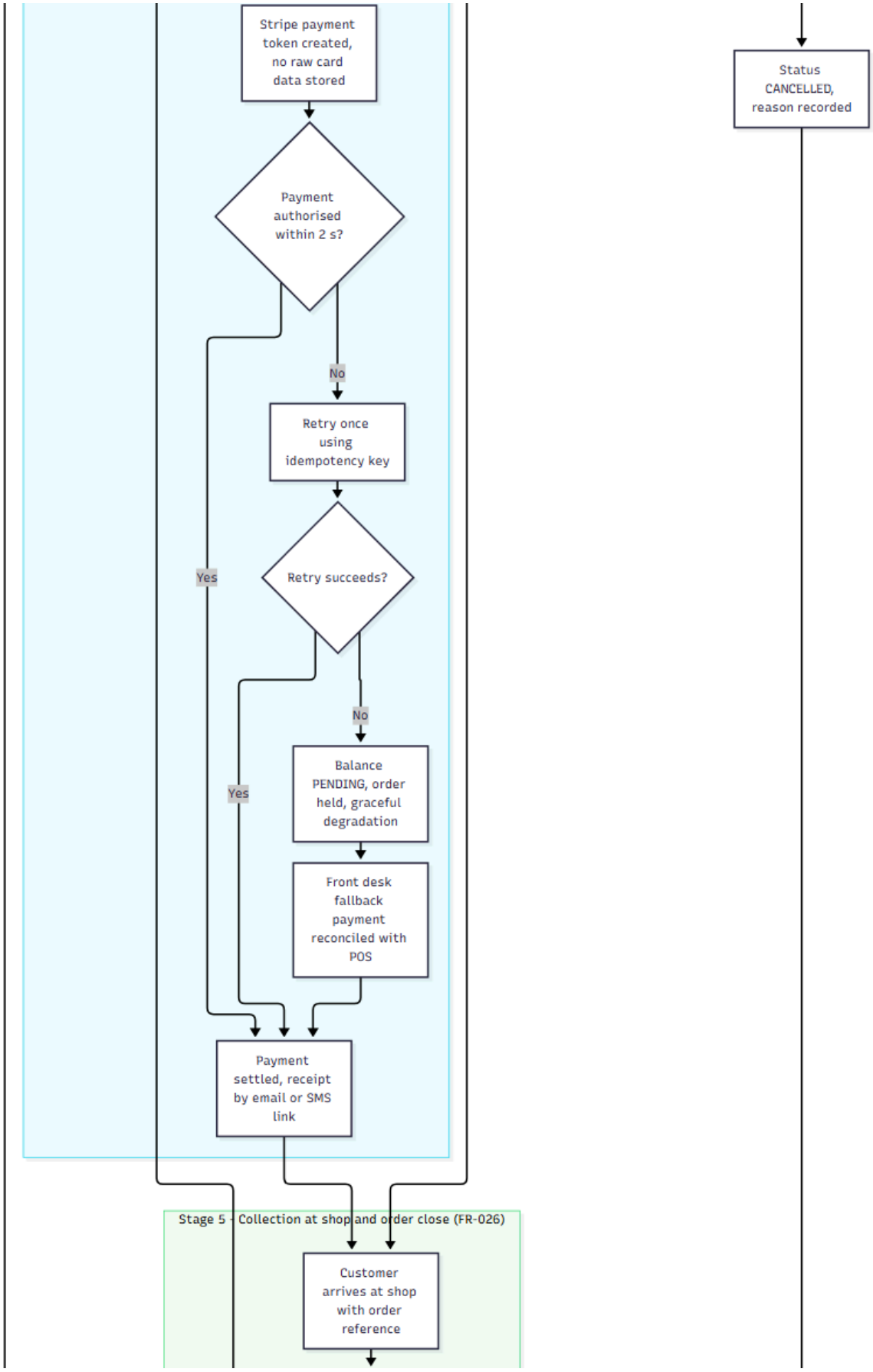


Figure 6: Process Diagram (part 6 of 8)

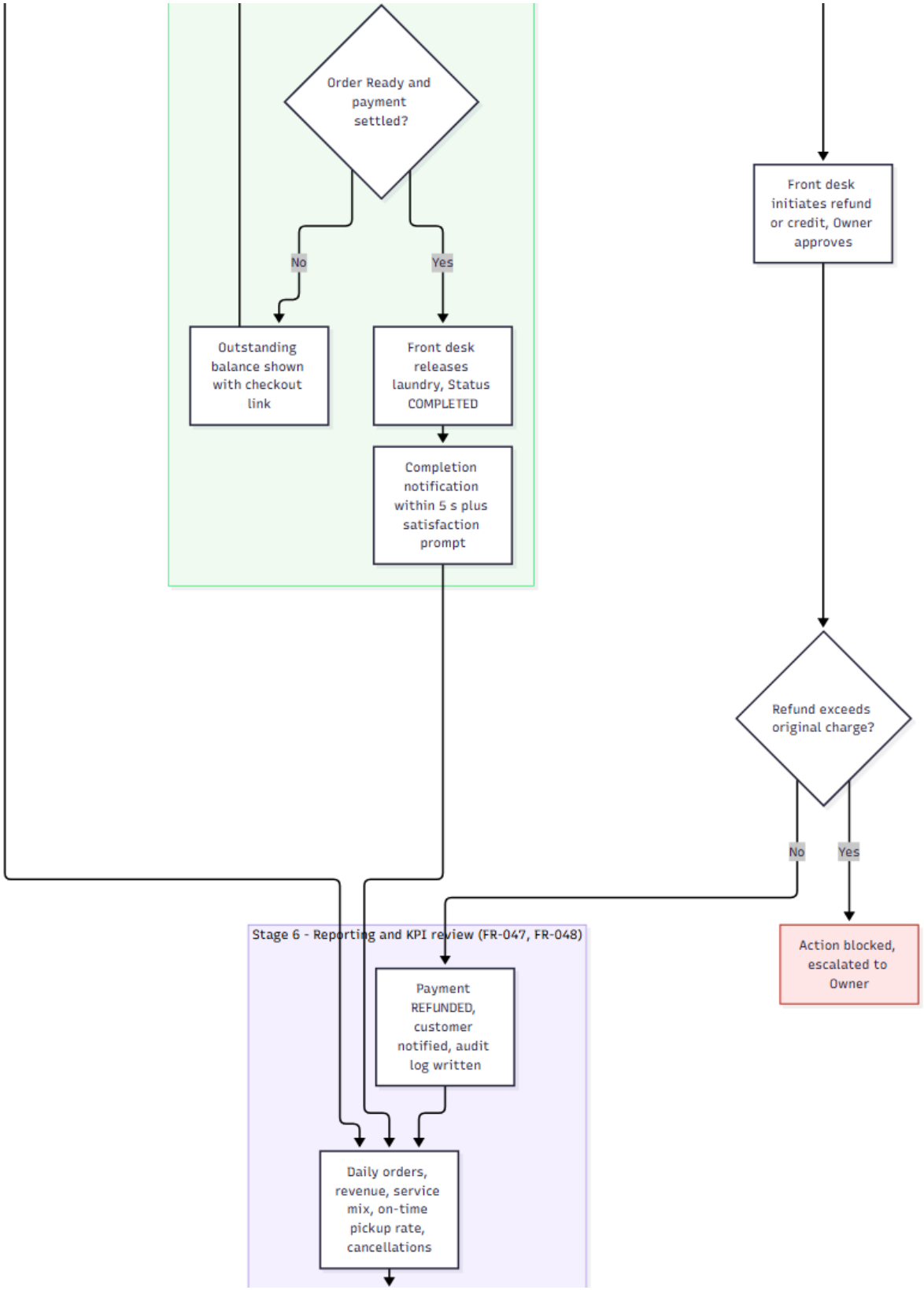


Figure 7: Process Diagram (part 7 of 8)



Figure 8: Process Diagram (part 8 of 8)

Order Status Lifecycle State Machine (FR-019)

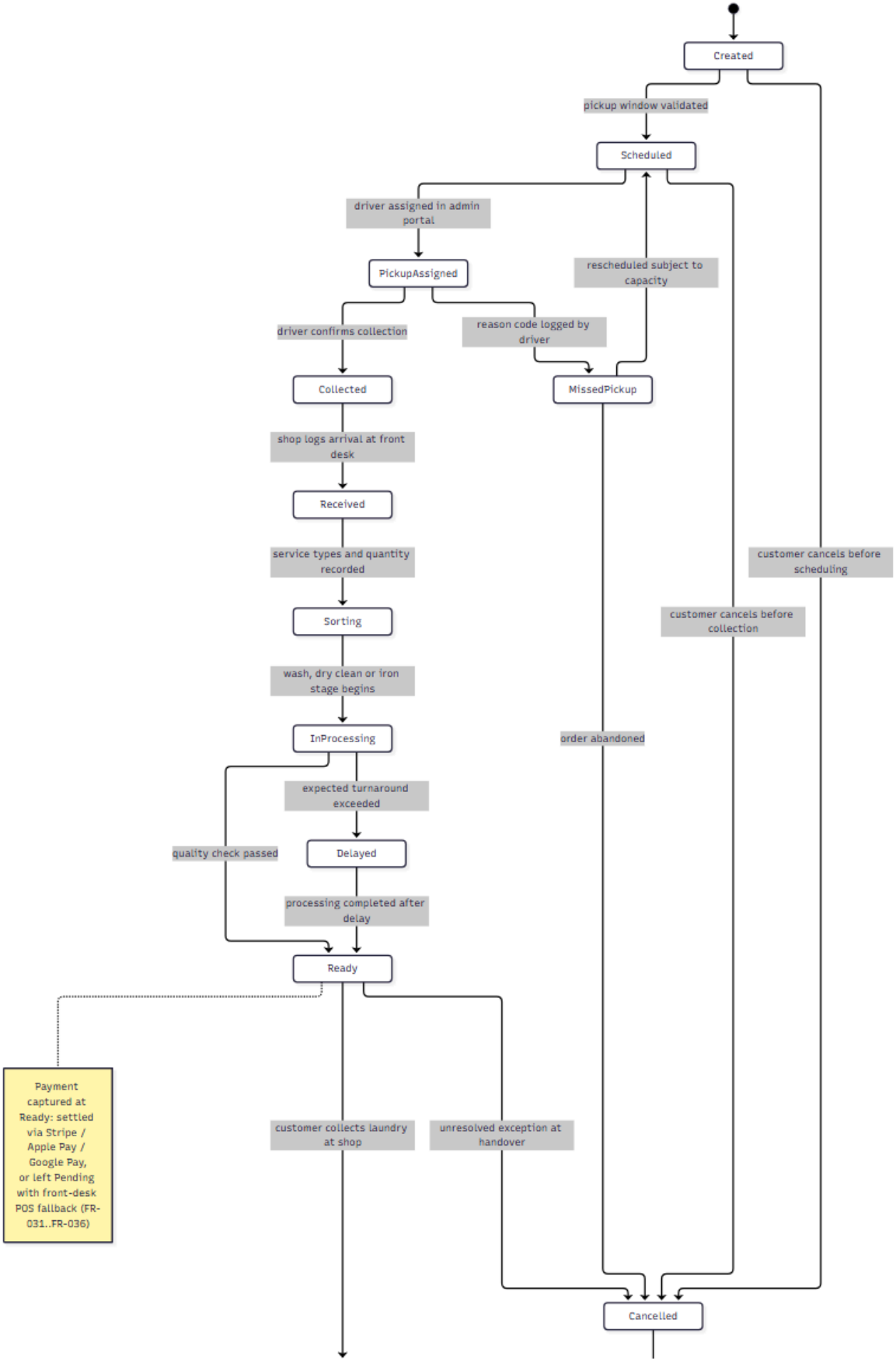


Figure 9: Order Status Lifecycle State Machine (FR-019) (part 1 of 2)

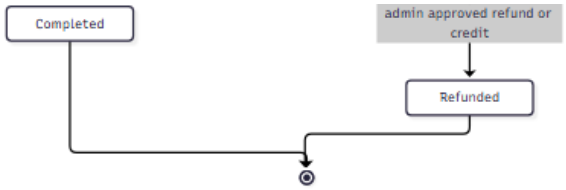


Figure 10: Order Status Lifecycle State Machine (FR-019) (part 2 of 2)

Status Change, Notification and Payment Sequence (UC-004, UC-006, UC-007, UC-010)

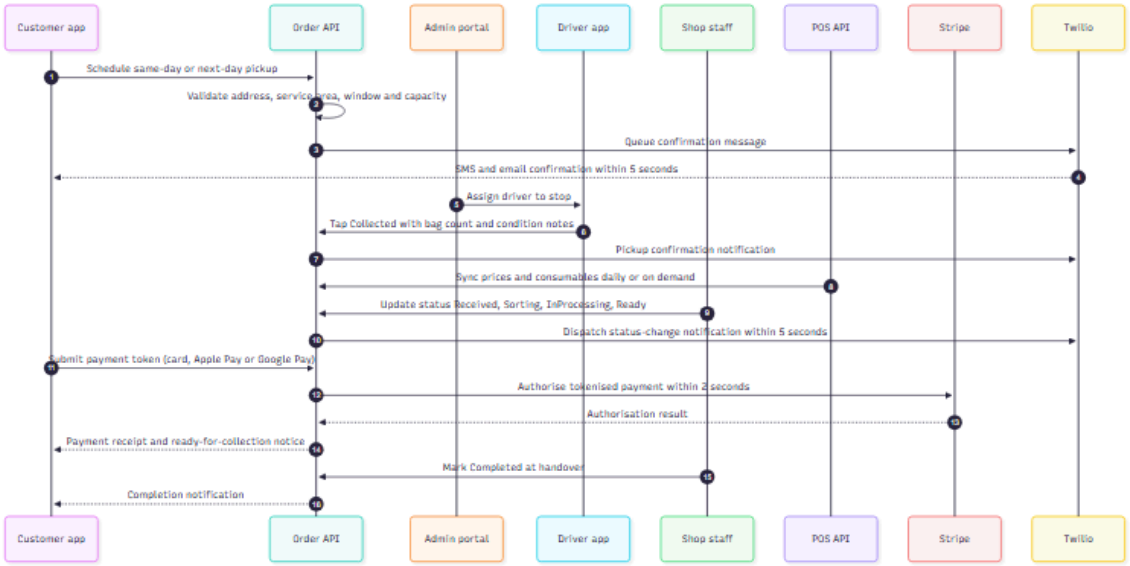


Figure 11: Status Change, Notification and Payment Sequence (UC-004, UC-006, UC-007, UC-010)

Exception Escalation Flow (missed pickup, damaged/missing item, payment failure)

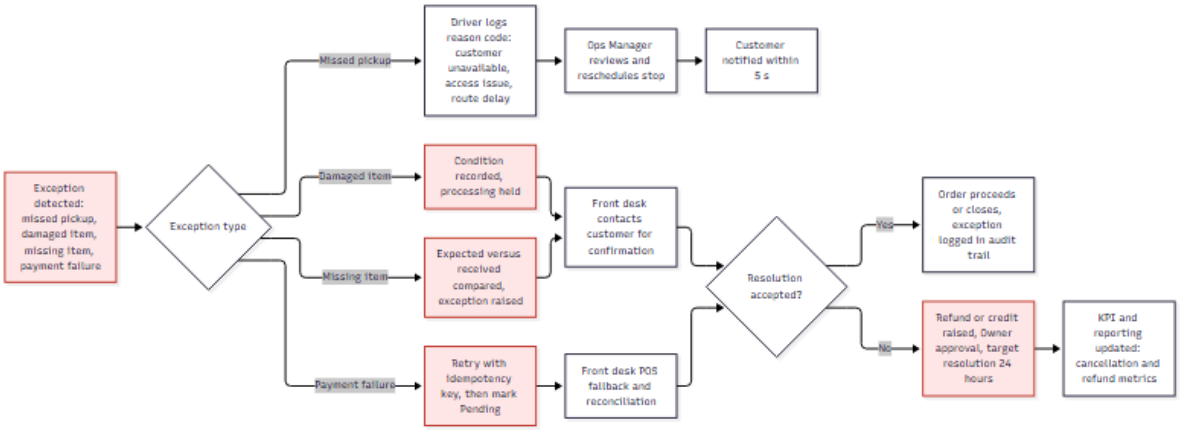


Figure 12: Exception Escalation Flow (missed pickup, damaged/missing item, payment failure)